International Institute of Information Technology Bhubaneswar
Subject name- Computer Vision
Mid-Semester Examination, September 2025
7^{th} Semester, B.Tech., IT Branch

F.M. – 30 | **Time – $1\frac{1}{2}$ Hours**

Answer any three including Question No.1
The figures in the right-hand margin indicate marks.

1. Answer All Questions [2×5]

   a. Define *Digital* image and *Analog* image. How does a digital image differ from an analog image in terms of representation and processing?
   b. What are *Binary* images and *Grayscale* images? Explain one key difference in how each image format stores pixel information.
   c. In *Grayscale* images, how is the range of intensity levels determined, and why is the total number of intensity levels (L) taken as 256?
   d. Differentiate between the two basic types of images: *Raster* and *Vector* images. Highlight one key difference in their structure and scalability.
   e. Explain the difference between *Lossy* and *Lossless* image compression. Provide one example of each type of compression technique.

2. (a) List and explain the various steps involved in digital image processing. Illustrate how these steps contribute to improving image quality and extracting meaningful information. How do computational cost and input quality affect the performance of DIP algorithms? [5]

   (b) Explain the working principle of infrared (IR) imaging and ultrasound imaging. Provide at least two practical applications for each modality in real-world scenarios. [5]

3. (a) Explain and compare the different intensity transformation techniques used for image enhancement [Negative, Logarithmic and Power-law (Gamma)] transformations. Discuss how each method affects image contrast and brightness with suitable examples. [5]

   (b) Explain the difference between Image Filtering and Image Warping. Also, discuss the role of image filtering in noise reduction, mentioning common types of noise encountered in digital images. [5]

4. (a) A point P(2,3) undergoes the following transformations sequentially: [10]
   (i) translation by (3, –2),
   (ii) rotation by 90° counter-clockwise about the origin, and
   (iii) scaling by 2 in x and 1 in y.

   (b) Find the final coordinates of P′ using homogeneous matrix multiplication.